

1-d bifurcations continued

$$\dot{x} = f(x; \lambda), \quad x, \lambda \in \mathbb{R}$$

$$f(x^*; \lambda^*) = \frac{\partial f}{\partial x}(x^*; \lambda^*) = 0$$

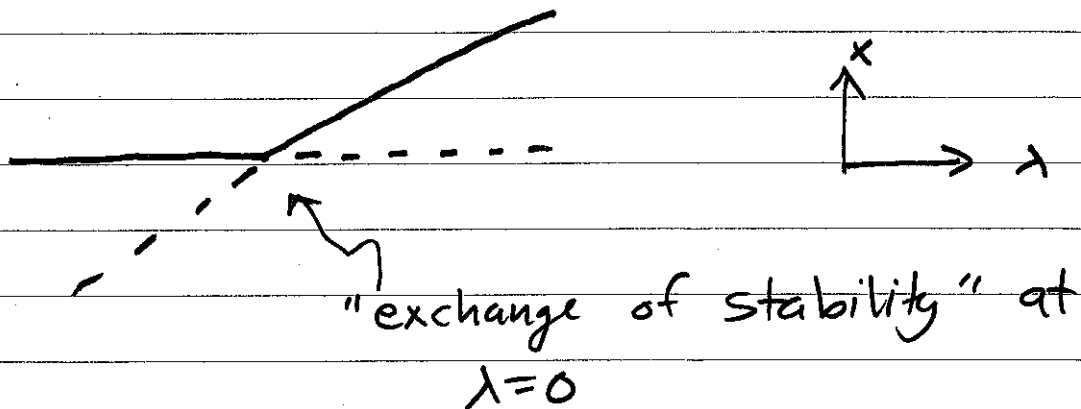
bifurcation
conditions"Transcritical Bifurcation"

$$\frac{\partial f}{\partial \lambda}(x^*; \lambda^*) = 0$$

can happen if there is a soln. that exists for all λ such as zero population
 $(x^* = 0) \Rightarrow \dot{x} = xg(x; \lambda) \Rightarrow \frac{\partial f}{\partial \lambda}(x^* = 0) = 0$

assume $\frac{\partial^2 f}{\partial x \partial \lambda}(x^*; \lambda^*) \neq 0$ & $\frac{\partial^2 f}{\partial x^2}(x^*; \lambda^*) \neq 0$

example: $\dot{x} = \lambda x - x^2 = x(\lambda - x)$



Pitchfork bifurcation: expected if there is a symmetry $x \rightarrow -x$

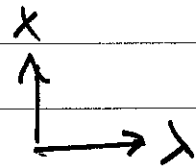
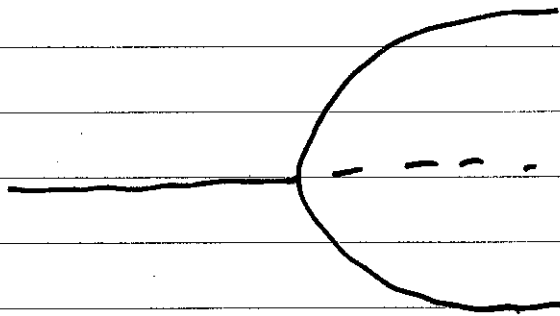
$$\dot{x} = x g(x^2, \lambda)$$

odd in $x \Rightarrow \frac{\partial f}{\partial x^2}(0, \lambda^*) = \frac{\partial f}{\partial \lambda}(0, \lambda^*) = 0$

assume $\frac{\partial^2 f}{\partial x \partial \lambda}(0, \lambda^*) \neq 0$ & $\frac{\partial^3 f}{\partial x^3}(0, \lambda^*) \neq 0$

Pitchfork example:

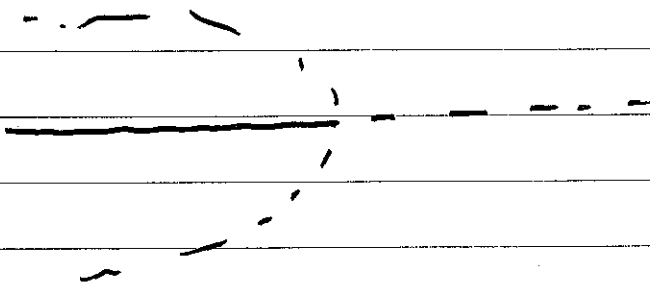
$$\dot{x} = x(\lambda - x^2) = \lambda x - x^3$$



"supercritical"
pitchfork

$$\dot{x} = x(\lambda + x^2) = \lambda x + x^3$$

"subcritical"
pitchfork



Pitchfork in terms of a potential $V(x)$

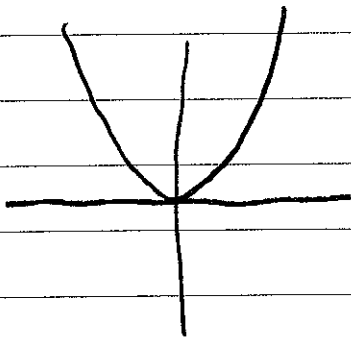
$$\frac{dx}{dt} = f(x), \quad x \in \mathbb{R}$$

$$= -\frac{dV}{dx}, \quad V = -\int f(x) dx$$

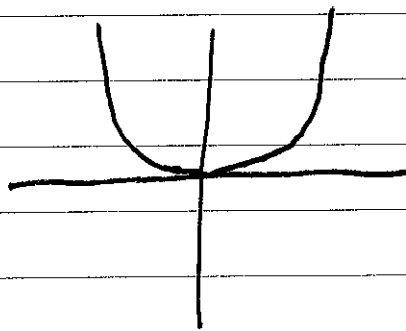
let $V(t) = V(x(t))$ where $\dot{x} = -\frac{dV}{dx}$

$$\Rightarrow \frac{dV}{dt} = \frac{dV}{dx} \frac{dx}{dt} = -\left(\frac{dV}{dx}\right)^2 \leq 0$$

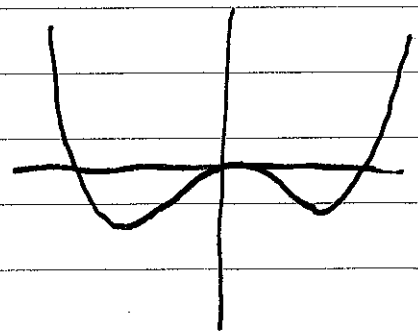
$$f = \lambda x - x^3 \Rightarrow V = -\frac{\lambda}{2} x^2 + \frac{1}{4} x^4$$



$\lambda < 0$



$\lambda = 0$



$\lambda > 0$

symmetric
double-well

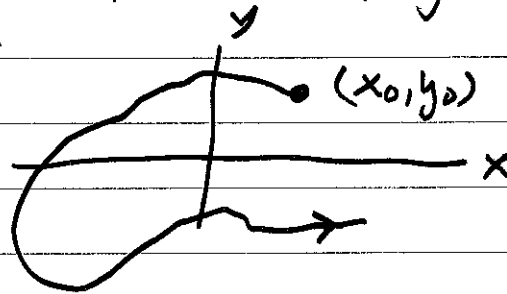
This leads to "spontaneous symmetry-breaking"

don't confuse $V(x)$ here with physics case
 acceleration $\rightarrow m\ddot{x} = -\frac{dV}{dx}$ vs $\dot{x} = -\frac{dV}{dx} \leftarrow$ velocity

Strogatz Part II: 2-D flows

$$\left. \begin{aligned} \dot{x} &= f(x, y) \\ \dot{y} &= g(x, y) \end{aligned} \right\} \begin{array}{l} \text{state space variables} \\ (x, y) \in \mathbb{R}^2 \end{array}$$

Solutions $(x(t), y(t))$ with $(x(0), y(0)) = (x_0, y_0)$ are t -parameterized trajectories in the phase plane:



phase plane

Start with a classification of linear, homogeneous system

$(x, y) = (0, 0)$ is a soln.

$$\left. \begin{aligned} \dot{x} &= ax + by \\ \dot{y} &= cx + dy \end{aligned} \right\} a, b, c, d \in \mathbb{R}$$

$$\underbrace{\begin{pmatrix} \dot{x} \\ \dot{y} \end{pmatrix}}_{\dot{\mathbf{x}}} = \underbrace{\begin{pmatrix} a & b \\ c & d \end{pmatrix}}_A \underbrace{\begin{pmatrix} x \\ y \end{pmatrix}}_{\mathbf{x}}$$

$$\dot{\mathbf{x}} = A\mathbf{x}$$

Lecture 5 p.5

This includes 2nd order eqns. of the form

$$\ddot{x} + \beta \dot{x} + \alpha x = 0$$

$$\begin{aligned}\dot{x} &= y \\ \dot{y} &= -\beta y - \alpha x\end{aligned}$$

$$A = \begin{pmatrix} 0 & 1 \\ -\alpha & -\beta \end{pmatrix}$$

$$(*) \quad \dot{X} = AX, \quad X \in \mathbb{R}^2$$

If we have 2 linearly independent solns $x_1(t)$ & $x_2(t)$, then we have the general soln. of (*):

$$X(t) = c_1 x_1(t) + c_2 x_2(t)$$

Initial condition determines c_1, c_2

$$X_0 = c_1 x_1(0) + c_2 x_2(0) = \begin{bmatrix} 1 & 1 \\ x_1(0) & x_2(0) \\ 1 & 1 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix}$$

invertible since
 $x_1(t)$ & $x_2(t)$
are linearly
independent

$$\Rightarrow \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ x_1(0) & x_2(0) \\ 1 & 1 \end{bmatrix}^{-1} X_0$$

$$\dot{X} = AX$$

($\dot{x} = \lambda x$ has soln. $x(t) = e^{\lambda t} x_0$)

Try similar form here: $X(t) = e^{\lambda t} V$,
where $V = \text{constant vector}$, which is the initial condition.

$$\left. \begin{array}{l} \frac{dX}{dt} = \lambda e^{\lambda t} V \\ AX = e^{\lambda t} AV \end{array} \right\} \Rightarrow \underbrace{\lambda V = AV}_{\substack{\lambda = \text{eigenvalue of } A \\ \& V = \text{eigenvector associated} \\ \text{with } \lambda}}$$

eigenvalue problem: $(A - \lambda I)V = 0$

has a nontrivial soln. provided
 $\text{Det}(A - \lambda I) = 0$, otherwise $V = 0$ is the unique soln.

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \quad A - \lambda I = \begin{pmatrix} a - \lambda & b \\ c & d - \lambda \end{pmatrix}$$

$$\text{Det}(A - \lambda I) = (a - \lambda)(d - \lambda) - bc$$

$$= \lambda^2 - \underbrace{(a + d)}_{\tau} \lambda + \underbrace{(ad - bc)}_{\Delta}$$

$$\tau = \text{Trace}(A), \quad \Delta = \text{Det}(A)$$

$$\lambda_{\pm} = \frac{-\tau \pm \sqrt{\tau^2 - 4\Delta}}{2}$$

$$\begin{cases} \Delta = \lambda_+ \lambda_- \\ \tau = \lambda_+ + \lambda_- \end{cases}$$

Note: λ_+ & λ_- may be complex: depends on $\text{sign}(\tau^2 - 4\Delta)$

